

Solutions to Quiz 9

Question 1

Suppose that $\mathbb{Q}[\sqrt{5}]$ is contained in a field extension K of $\mathbb{Q}$ which has dimension $[K : \mathbb{Q}] = 6$ as a vector space over $\mathbb{Q}$. What is the characteristic polynomial of $\sqrt{5} - 1$ as an element of K ?

Solution:

We note that K is a vector space of dimension 3 over $\mathbb{Q}[\sqrt{5}]$. Let $U = \{\alpha_1, \alpha_2, \alpha_3\}$ be a basis.

The set $T = \{1, \sqrt{5}\}$ is a basis for $\mathbb{Q}[\sqrt{5}]$ over $\mathbb{Q}$.

If we take $S = \{\alpha_1, \alpha_1\sqrt{5}, \alpha_2, \alpha_2\sqrt{5}, \alpha_3, \alpha_3\sqrt{5}\}$ as a basis for K over $\mathbb{Q}$. We see that the matrix $M_S(\sqrt{5})$ is a block diagonal matrix with blocks $\begin{pmatrix} 0 & 5 \\ 1 & 0 \end{pmatrix}$.

It follows that $M_S(\sqrt{5} - 1)$ is a block diagonal matrix with blocks $\begin{pmatrix} -1 & 5 \\ 1 & -1 \end{pmatrix}$.

Thus, its characteristic polynomial is the third power of the characteristic polynomial of this matrix. We calculate the latter as $(X + 1)^2 - 5 = X^2 + 2X - 4$.

Thus, the answer is $(X^2 + 2X - 4)^3$.

Question 2

Suppose that u is a unit in the ring of algebraic integers of a finite field extension of K over $\mathbb{Q}$. What are the possible values for the norm $\text{Nm}_{K/\mathbb{Q}}(u)$?

Solution:

We have seen that there is a basis $S = \{\alpha_1, \dots, \alpha_n\}$ of K over $\mathbb{Q}$ so that the ring of integers $\mathcal{O}_K$ is *precisely* integer linear combinations of the α_i . Equivalently, $M_S(\alpha)$ is in $M_n(\mathbb{Z})$ if and only if α lies in $\mathcal{O}_K$.

Since u is in $\mathcal{O}_K$, we see that $M_S(u)$ is in $M_n(\mathbb{Z})$. Since u is a unit, $1/u$ is also in $\mathcal{O}_K$. Thus, $M_S(1/u) = M_S(u)^{-1}$ is also in $M_n(\mathbb{Z})$. This means that $\det M_S(u)$ and $\det M_S(u)^{-1}$ are integers that are inverses of each other. This means that $\det M_S(u) = \pm 1$. Hence, these are the possible values of $\text{Nm}_{K/\mathbb{Q}}(u)$.